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Flex
Automotive Power Electronics Team
Stuttgart

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Application: Working Student / Master Student Internship – Power Electronics Simulation & Converter Analysis

Dear Automotive Power Electronics Team,

Onboard chargers and DC/DC converters only reach the road if their designs are verified under worst-case conditions long before hardware exists. That is the work in your Power Electronics Simulation & Converter Analysis role, and it is the part of power electronics I most want to build experience in. I am an M.Sc. Electrical Engineering student at the University of Stuttgart, with coursework in power electronics, power systems and battery modeling.

Your posting asks for initial SPICE experience, and I have it. I am PSpice certified (TVS India), and circuit simulation is one of my core skills. During my one-year internship at FocalSystech, I designed and validated electronic circuits and evaluated components against technical requirements. I also automated testing and performance analysis in Python and embedded C, which gives me a head start on the simulation-workflow automation you mention.

In a university team project on battery cell modeling, I analysed how electrical, thermal, aging and manufacturing tolerances spread cell behaviour, and implemented and debugged statistical parameter sampling in the simulation model. This is not the same as a formal worst case circuit analysis, but it trained the same habit: asking what happens at the edges of the tolerance range, and checking that the model is right before trusting its result.

I should be honest about the gaps. I have not yet worked on onboard chargers or resonant converters and have used PSpice rather than LTspice. I learn tools quickly, and I would value working with your engineers on real automotive platforms. My English is fluent. My German is at A2 and I am preparing for the telc B1 exam; I would gladly use the German courses Flex offers.

I can work up to 20 hours per week alongside my studies and am happy to agree a start date. I look forward to discussing how I can support your team.

Kind regards,

Vineeth Raghav Gururajan

Enclosures: CV, proof of enrolment, certificates